18 Bayesian Analyis of a Numerical Variable

- When the distribution of the measured variable is symmetric and unimodal, the population mean μ is often the main parameter of interest. However, the population SD σ also plays an important role.
- We have previously assumed that the measured numerical variable follows a conditional Normal(μ, σ) distribution. That is, we assumed a Normal-based likelihood function.
- However, the assumption of Normality is not always justified, even when the distribution of the measured variable is symmetric and unimodal.

Example 18.1. Assume that birthweights (grams) of human babies follow a Normal distribution with unknown mean μ and unknown standard deviation σ. (1 pound $\approx$ 454 grams.)

1. Considering μ in isolation, what seems like a reasonable prior distribution for μ?

2. What does the parameter σ represent? What is a reasonable prior distribution for σ?

3. Assume a Normal(3400, 100) prior distribution for μ, a Normal(600, 200) prior distribution for σ, and that μ and σ are independent. Explain how you could use simulation to approximate the prior predictive distribution of birthweights. Run the simulation and summarize the results. Does the choice of prior seem reasonable?

Example 18.2. Section 18.1.2 summarizes data on a random sample[1] of 1000 live births in the U.S. in 2001.

1. Inspect the sample data; does it seem reasonable to assume birthweights follow a Normal distribution?

2. Regardless of your answer to the previous question, continue to assume the conditional Normal(μ, σ) model for birthweights, with prior distributions as in the previous example. Use `brms` to find the posterior distribution. Briefly describe the posterior distribution.

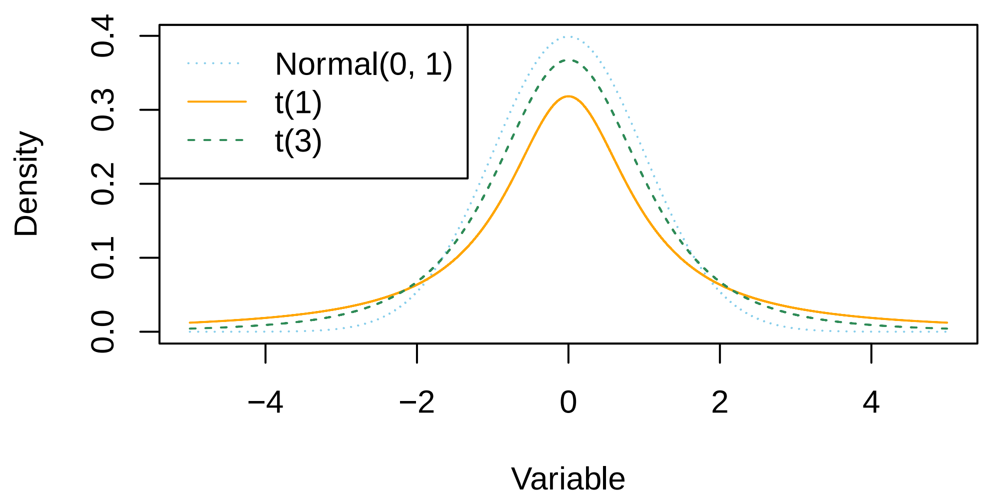
3. How could you use simulation to approximate the posterior predictive distribution of birthweights? Run the simulation and find a 99% posterior prediction interval.

4. What percent of values in the observed *sample* fall outside the prediction interval? What does that tell you?

5. Run a posterior predictive check. Does it indicate any problems?

[1]There are about 4 million live births in the U.S. per year. The data is available at the CDC website. We're only using a random sample to cut down on computation time.

- If the observed data is relatively unimodal and symmetric but has more extreme values than can be accommodated by a Normal-based likelihood, a t-distribution or other distribution with heavy tails can be used to model the likelihood.
- For "Student" t-distributions, the *degrees of freedom* parameter $\nu \geq 1$ controls how heavy the tails are.
  - When ν is small, the tails are much heavier than for a Normal distribution, leading to a higher frequency of extreme values.
  - As ν increases, the tails get lighter and a t-distribution gets closer to a Normal distribution.
  - For ν greater than 30 or so, there is very little different between a t-distribution and a Normal distribution except in the extreme tails.
- The degrees of freedom parameter ν is sometimes referred to as the "Normality parameter", with larger values of ν indicating a population distribution that is closer to Normal.



Example 18.3. Continuing the birthweight example, we'll now model the distribution of birthweights with a conditional $t(\mu, \sigma, \nu)$ distribution.

1. How many parameters is the likelihood function based on? What are they?

2. What does assigning a prior distribution to the Normality parameter ν represent?

3. Software packages like `brms` will choose a prior for you if you don't specify one. (Basically, the software will run some prior predictive tuning to find a choice of prior that results in a reasonable scale for the response variable.) Change `family` from `gaussian` to `student` and fit the model with `brm`. Run `prior_summary()`; what prior distribution did `brms` assume for `nu`?

4. Consider the posterior distribution for ν. Based on this posterior distribution, is it plausible that birthweights follow a Normal distribution?

5. Consider the posterior distribution for σ. What seems strange about this distribution? (Hint: consider the sample SD.)

6. Using the `brms` output, create a plot of the posterior distribution of $\sigma\sqrt{\frac{\nu}{\nu-2}}$. Does this posterior distribution of the population standard deviation seem more reasonable in light of the sample data?

- The standard deviation of a Normal(μ, σ) distribution is σ.

- However, the standard deviation of a $t(\mu, \sigma, \nu)$ distribution is not σ; rather it is $\sigma\sqrt{\frac{\nu}{\nu-2}} > \sigma$ (as long as $\nu > 2$; if $\nu \leq 2$ then the SD is infinite/undefined).
- When ν is large, $\sqrt{\frac{\nu}{\nu-2}} \approx 1$, and so the standard deviation is approximately σ. However, it can make a difference when ν is small.

Example 18.4. Continuing the previous example.

1. How could you use simulation to approximate the posterior predictive distribution of birthweights? Run the simulation and find a 99% posterior prediction interval. How does it compare to the predictive interval from the model with the Normal likelihood?

2. Run a posterior predictive check. Which model seems to fit the data better: the one based on the Normal likelihood or the t likelihood?

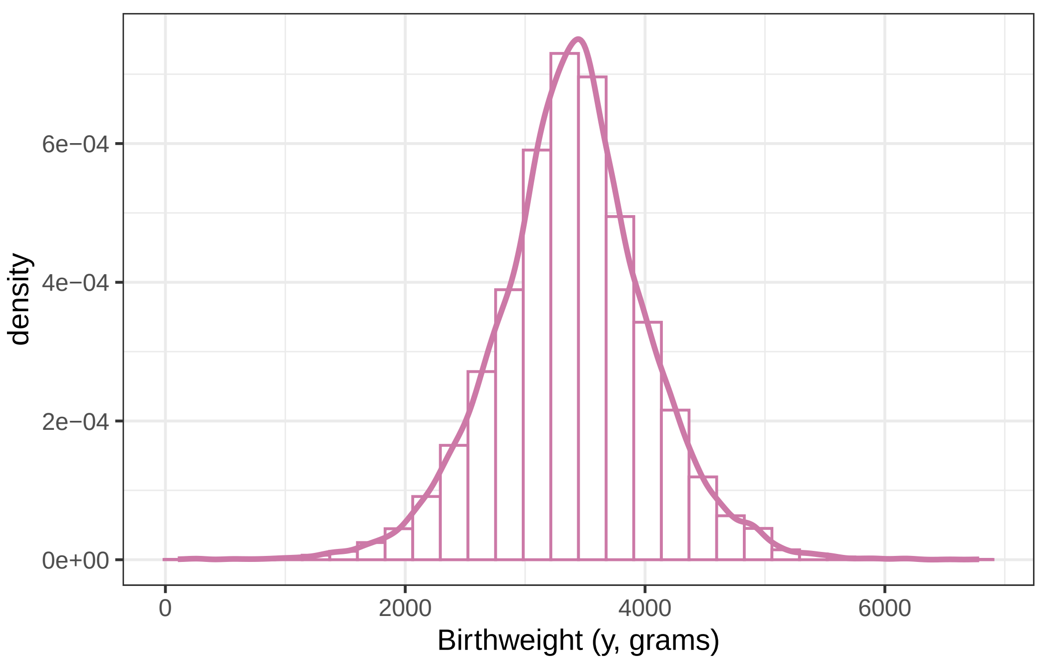
- While we do want a model that fits the data well, we also do not want to risk overfitting the data.
- In particular, we do not want a few extreme outliers to unduly influence the model.
- However, likelihood models based on distributions, like t, with heavier tails than Normal often provide more flexible and robust models.
- Moreover, accommodating the tails often improves the fit in the center of the distribution too.

- In models with multiple parameters, there can be dependencies between parameters, so interpreting the marginal posterior distribution of any single parameter can be difficult.
- It is often more helpful to consider predictive distributions, which accounts for the joint distribution of all parameters.
- Interpreting predictive distributions is often more intuitive since predictive distributions live on the scale of the measured variable.

18.1 Notes

18.1.1 Prior predictive distribution (Normal likelihood)

```
n_rep = 10000

mu_sim = rnorm(n_rep, 3400, sd = 100)

sigma_sim = pmax(0.001, rnorm(n_rep, 600, 200))

y_sim = rnorm(n_rep, mu_sim, sigma_sim)

ggplot(data.frame(y_sim), aes(x = y_sim)) +
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["prior_predict"],
    fill = "white"
  ) +
  geom_density(col = bayes_col["prior_predict"], linewidth = 1) +
  labs(x = "Birthweight (y, grams)") +
  theme_bw()
```



18.1.2 Sample data

```r
data = read_csv("birthweight.csv")
```

```
Rows: 1000 Columns: 1
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (1): birthweight

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(data)
```

```
# A tibble: 6 x 1
  birthweight
        <dbl>
1        3033
2        3147
3        3175
4        3005
5        2863
6        2665
```

```r
summary(data$birthweight)
```

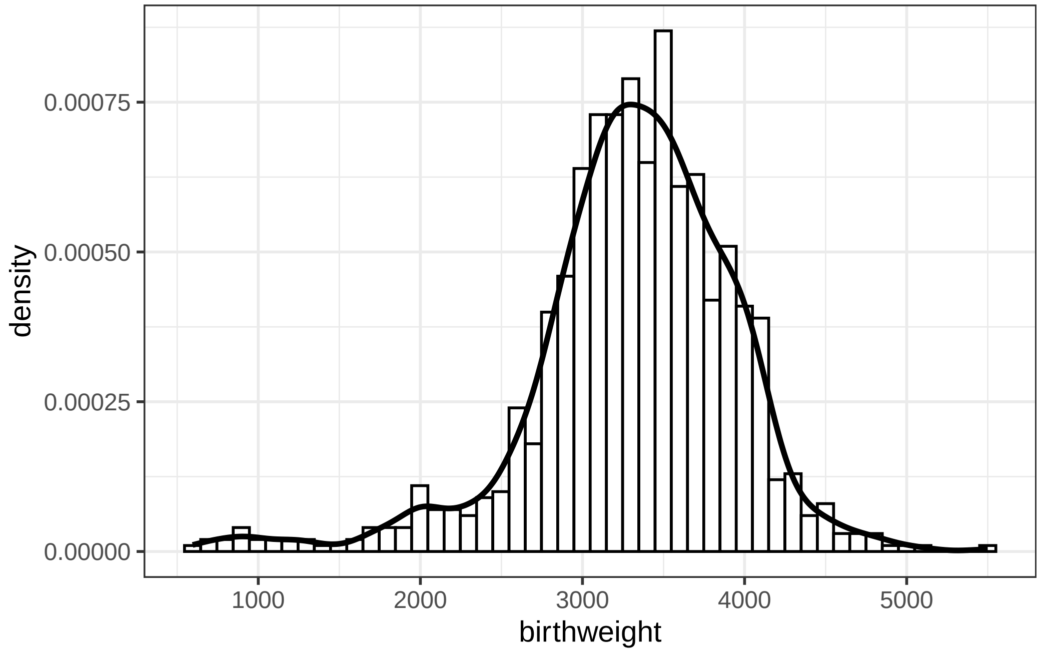
```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    595    3005    3350    3315    3714    5500
```

```r
sd(data$birthweight)
```

```
[1] 631.3303
```

```r
ggplot(data, aes(x = birthweight)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 50,
    col = "black",
```

```
    fill = "white"
  ) +
  geom_density(linewidth = 1) +
  theme_bw()
```



18.1.3 Posterior distribution (Normal likelihood)

```
library(brms)
```

```
Loading required package: Rcpp
```

```
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following objects are masked from 'package:tidybayes':

    dstudent_t, pstudent_t, qstudent_t, rstudent_t

The following object is masked from 'package:stats':

    ar
```

```
library(tidybayes)
library(bayesplot)
```

```
This is bayesplot version 1.14.0

- Online documentation and vignettes at mc-stan.org/bayesplot

- bayesplot theme set to bayesplot::theme_default()

   * Does _not_ affect other ggplot2 plots

   * See ?bayesplot_theme_set for details on theme setting


Attaching package: 'bayesplot'

The following object is masked from 'package:brms':

    rhat
```

```
fit <- brm(
  data = data,
  family = gaussian,
  birthweight ~ 1,
  prior = c(
    prior(normal(3400, 100), class = Intercept),
    prior(normal(600, 200), class = sigma)
  ),
  iter = 3500,
  warmup = 1000,
  chains = 4,
  refresh = 0
)
```

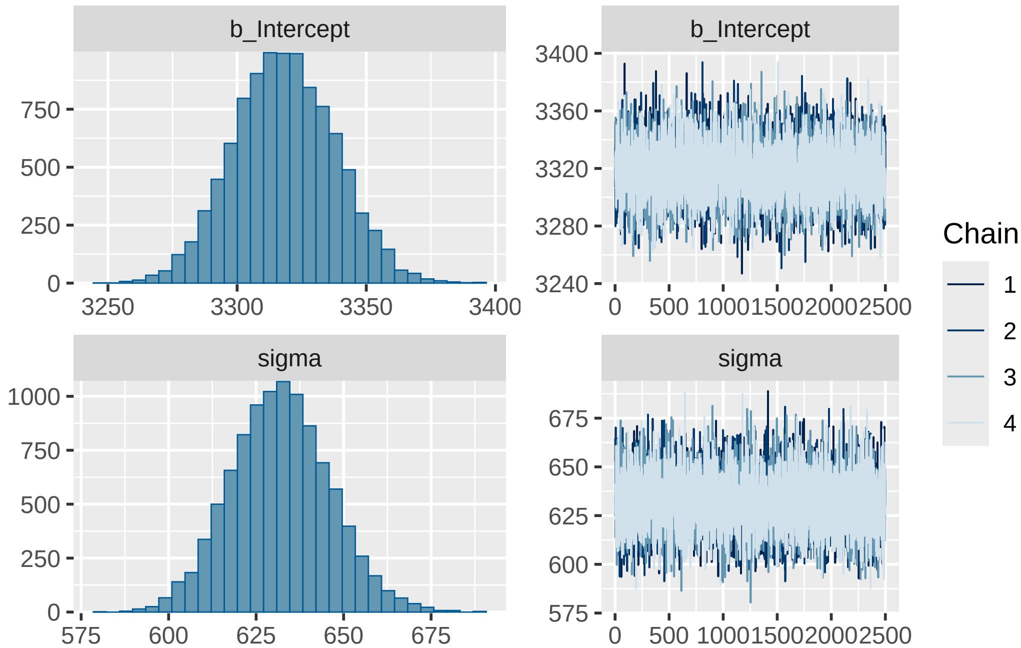
```
Compiling Stan program...
```

```
Start sampling
```

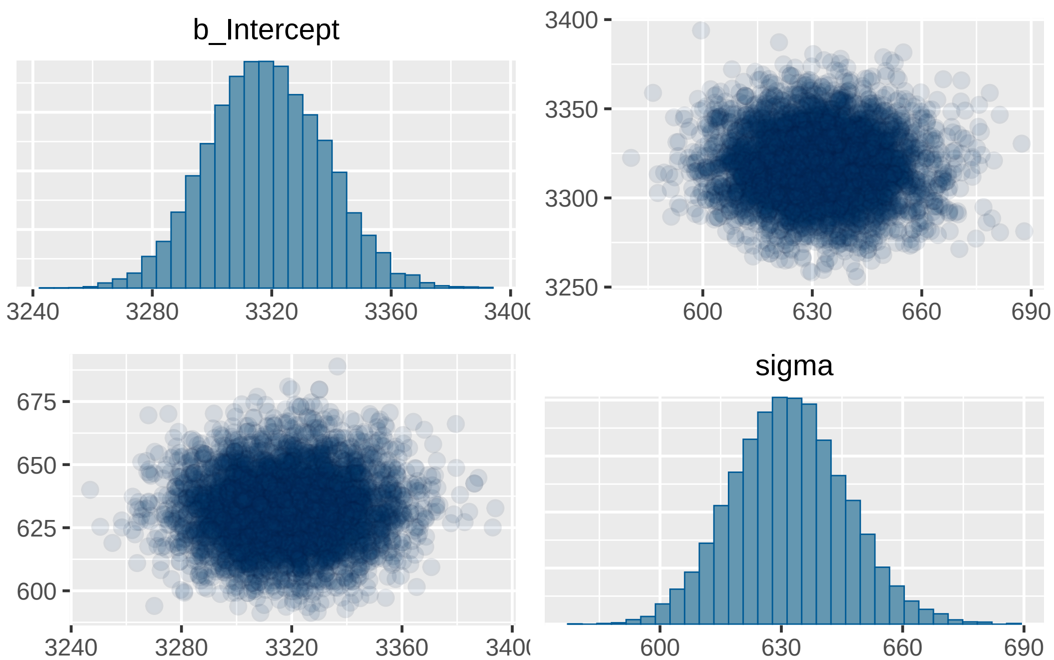
```r
summary(fit)
```

```
 Family: gaussian
  Links: mu = identity
Formula: birthweight ~ 1
   Data: data (Number of observations: 1000)
  Draws: 4 chains, each with iter = 3500; warmup = 1000; thin = 1;
         total post-warmup draws = 10000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept  3318.35     19.58  3280.84  3356.84 1.00     8676     7120

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   632.02     14.12   604.74   660.52 1.00     8646     6794

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit)
```




```
pairs(fit, off_diag_args = list(alpha = 0.1))
```



.chain	.iteration	.draw	mu	sigma
1	1	1	3328.840	664.5711
1	2	2	3349.764	652.9714
1	3	3	3352.722	657.4792
1	4	4	3284.833	659.3546
1	5	5	3325.805	645.9530
1	6	6	3315.574	613.5895
1	7	7	3354.471	621.3152
1	8	8	3312.937	654.0913
1	9	9	3310.302	634.6238
1	10	10	3317.561	631.4318

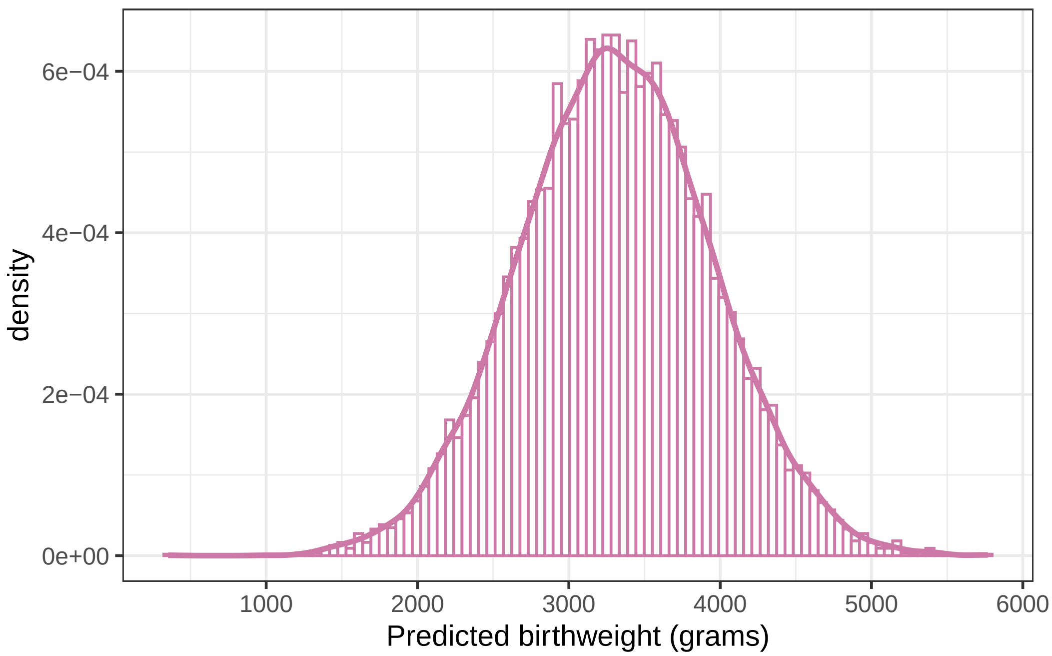
```r
posterior = fit |>
  spread_draws(b_Intercept, sigma) |>
  rename(mu = b_Intercept)

posterior |> head(10) |> kbl() |> kable_styling()
```

18.1.4 Posterior predictive distribution (Normal likelihood)

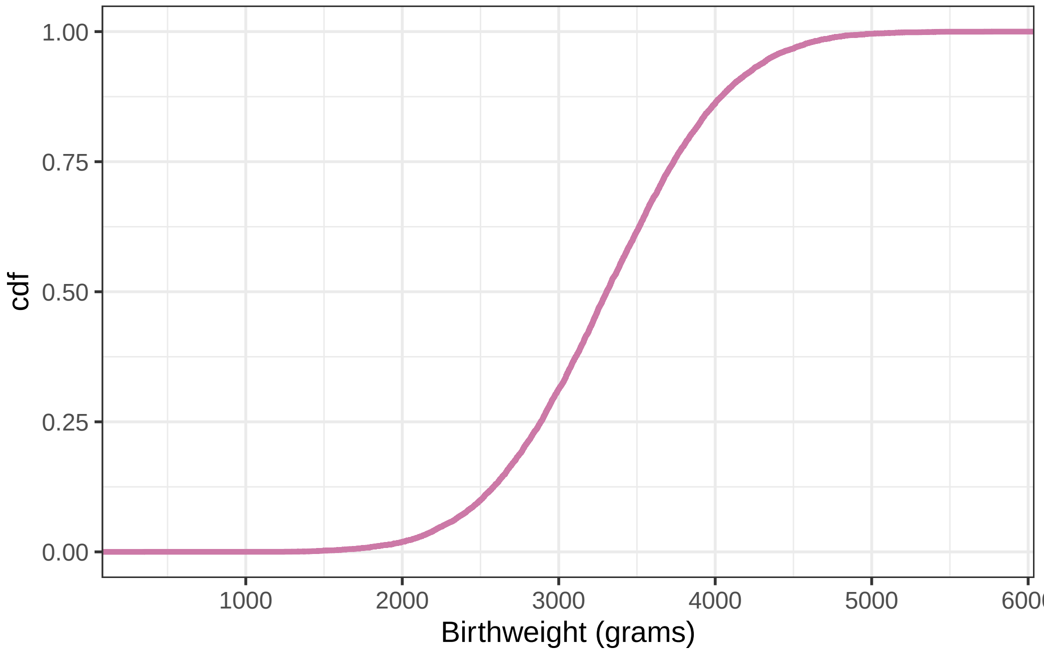
```r
n_rep = nrow(posterior)

posterior <- posterior |>
  mutate(y_sim = rnorm(n_rep, mean = posterior$mu, sd = posterior$sigma))

posterior |> head(10) |> kbl() |> kable_styling()
```

```r
ggplot(posterior, aes(x = y_sim)) +
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["posterior_predict"],
    fill = "white",
    bins = 100
  ) +
  geom_density(linewidth = 1, col = bayes_col["posterior_predict"]) +
  labs(x = "Predicted birthweight (grams)") +
  theme_bw()
```

.chain	.iteration	.draw	mu	sigma	y_sim
1	1	1	3328.840	664.5711	3315.678
1	2	2	3349.764	652.9714	2932.229
1	3	3	3352.722	657.4792	3919.526
1	4	4	3284.833	659.3546	2395.149
1	5	5	3325.805	645.9530	3412.514
1	6	6	3315.574	613.5895	3390.155
1	7	7	3354.471	621.3152	3159.211
1	8	8	3312.937	654.0913	2666.111
1	9	9	3310.302	634.6238	3061.508
1	10	10	3317.561	631.4318	3688.574




```
# plot the cdf
ggplot(posterior, aes(x = y_sim)) +
  stat_ecdf(col = bayes_col["prior_predict"], linewidth = 1) +
  labs(x = "Birthweight (grams)", y = "cdf") +
  theme_bw()
```




```r
quantile(posterior$y_sim, c(0.005, 0.995))
```

```
    0.5%    99.5%
1650.844 4955.230
```

```r
sum(data$birthweight < quantile(posterior$y_sim, 0.005)) /
  length(data$birthweight)
```

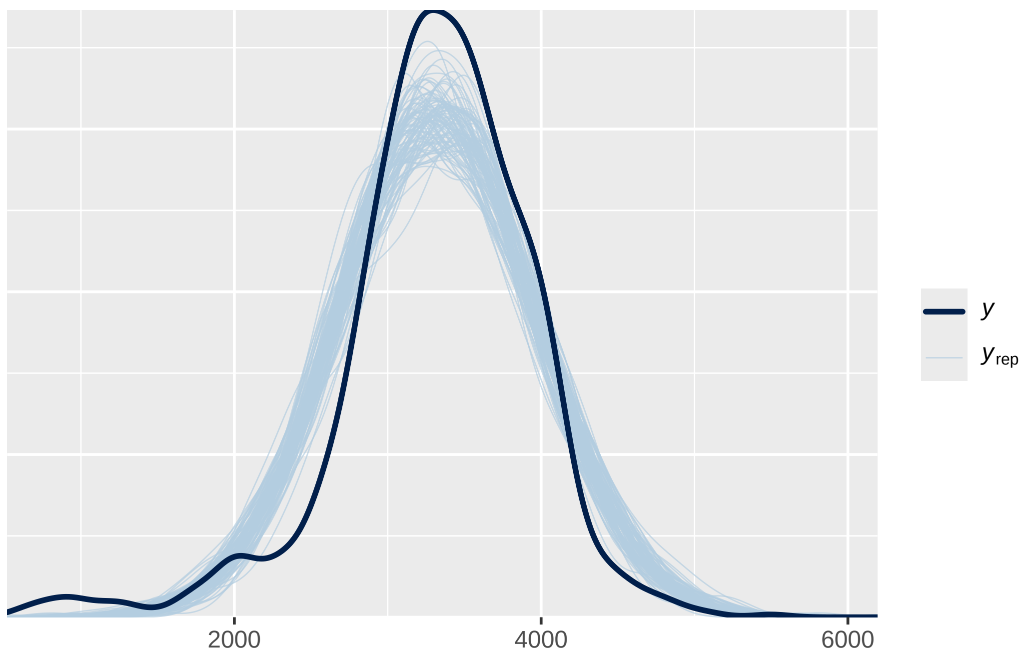
```
[1] 0.02
```

```r
sum(data$birthweight > quantile(posterior$y_sim, 0.995)) /
  length(data$birthweight)
```

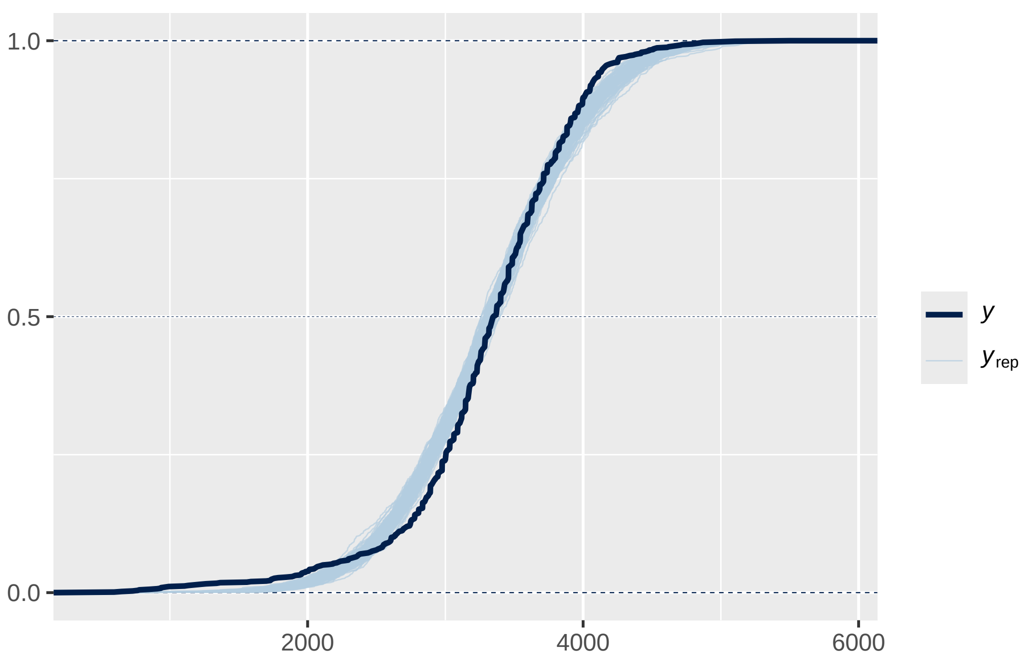
```
[1] 0.003
```

18.1.5 Posterior predictive checking (Normal likelihood)

```r
pp_check(fit, ndraw = 100)
```




```r
pp_check(fit, ndraw = 100, type = "ecdf_overlay")
```



18.1.6 Posterior distribution (t likelihood)

```r
fit <- brm(
  data = data,
  family = student,
  birthweight ~ 1,
  prior = c(
    prior(normal(3400, 100), class = Intercept),
    prior(normal(600, 200), class = sigma)
  ),
  iter = 3500,
  warmup = 1000,
  chains = 4,
  refresh = 0
)
```

```
Compiling Stan program...

Start sampling
```

```r
prior_summary(fit)
```

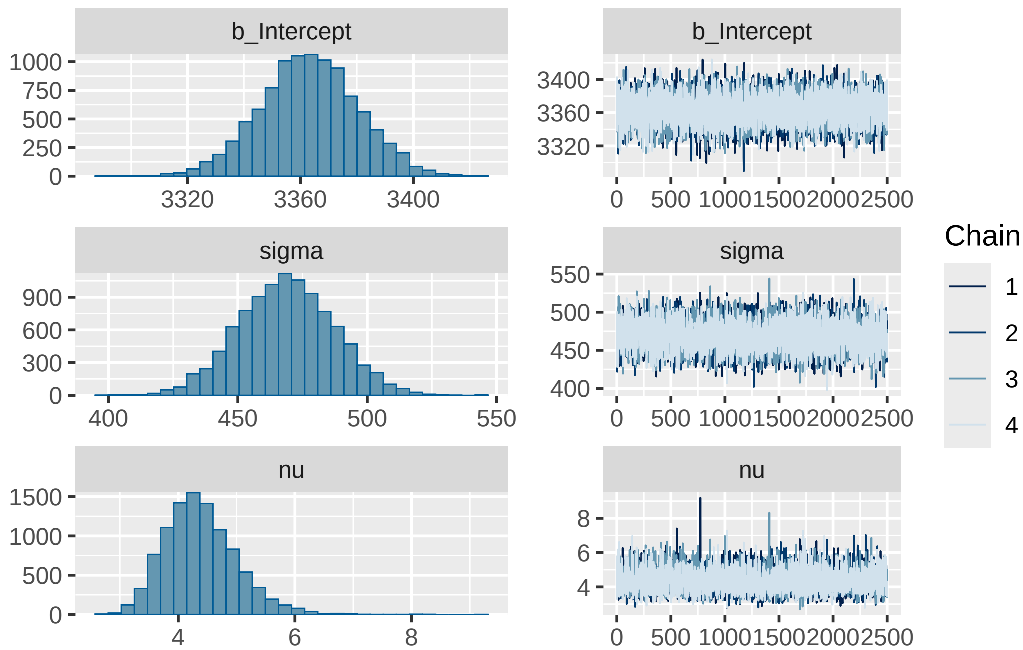
```
             prior     class coef group resp dpar nlpar lb ub tag  source
 normal(3400, 100) Intercept                                         user
     gamma(2, 0.1)        nu                             1        default
   normal(600, 200)     sigma                             0           user
```

```r
summary(fit)
```

```
 Family: student
  Links: mu = identity
Formula: birthweight ~ 1
   Data: data (Number of observations: 1000)
  Draws: 4 chains, each with iter = 3500; warmup = 1000; thin = 1;
         total post-warmup draws = 10000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept  3363.12     17.25  3328.99  3396.70 1.00     6746     5891
```

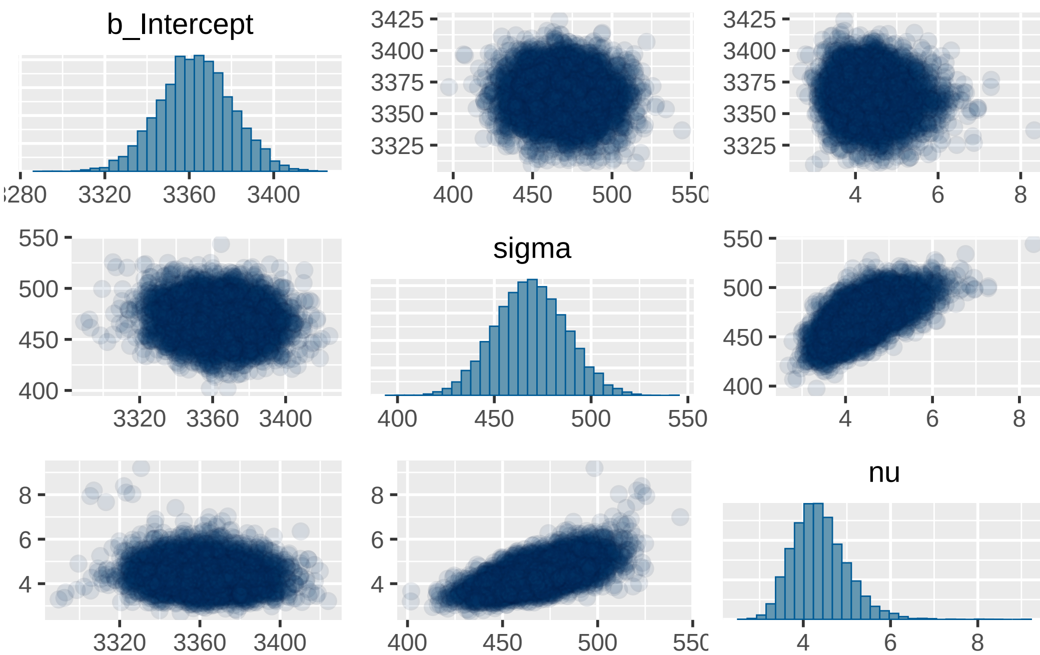
```
Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   468.82     18.36   433.30   504.96 1.00     5874     6042
nu        4.38      0.62     3.35     5.76 1.00     5959     6595

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit)
```




```r
pairs(fit, off_diag_args = list(alpha = 0.1))
```

.chain	.iteration	.draw	mu	sigma	nu	sd
1	1	1	3342.971	479.0871	4.288737	655.8148
1	2	2	3342.547	455.6609	4.325902	621.4188
1	3	3	3379.871	486.4239	4.604530	646.7597
1	4	4	3358.972	457.4084	4.224186	630.3621
1	5	5	3364.930	471.2112	3.824577	682.2228
1	6	6	3365.897	454.1934	3.628184	678.0064
1	7	7	3359.967	490.6955	4.884161	638.5536
1	8	8	3375.590	462.9417	4.380166	628.0120
1	9	9	3359.703	447.5173	3.919852	639.4562
1	10	10	3351.785	473.9757	4.294552	648.4344

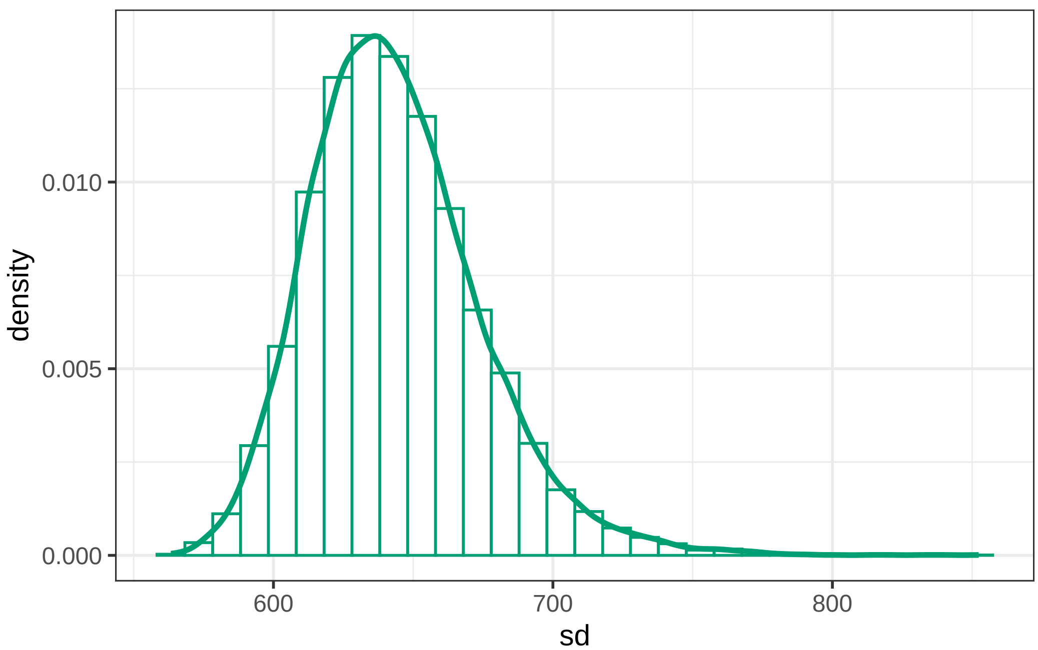



```r
posterior = fit |>
  spread_draws(b_Intercept, sigma, nu) |>
  rename(mu = b_Intercept) |>
  mutate(sd = sigma * sqrt(nu / (nu - 2)))

posterior |> head(10) |> kbl() |> kable_styling()
```

```r
posterior |>
  ggplot(aes(x = sd)) +
```

```
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["posterior"],
    fill = "white"
  ) +
  geom_density(linewidth = 1, col = bayes_col["posterior"]) +
  theme_bw()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

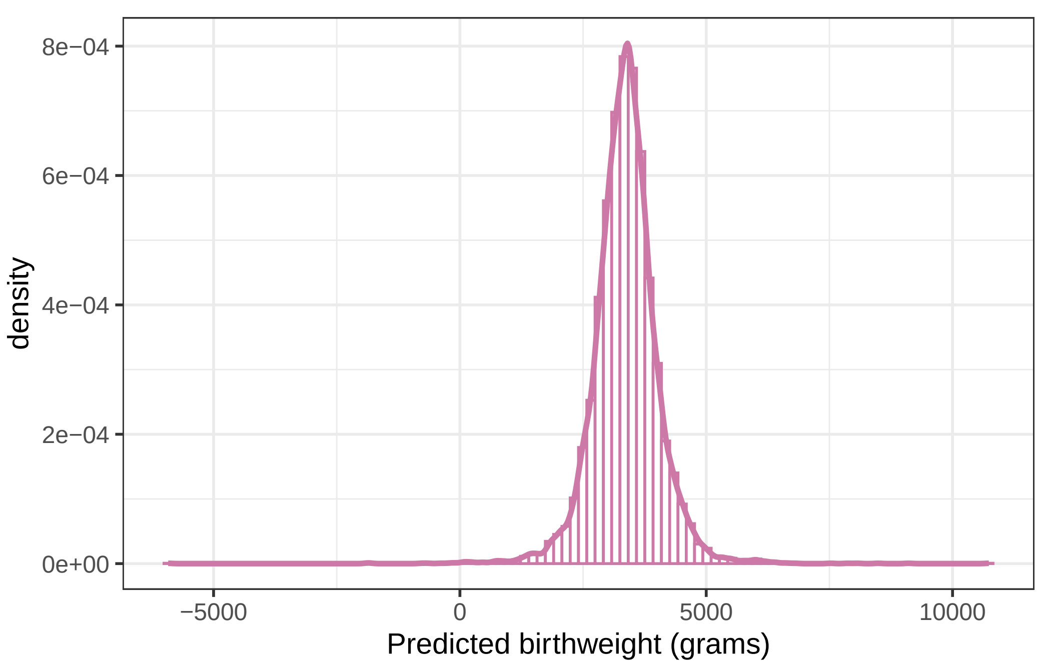


18.1.7 Posterior predictive distribution (t likelihood)

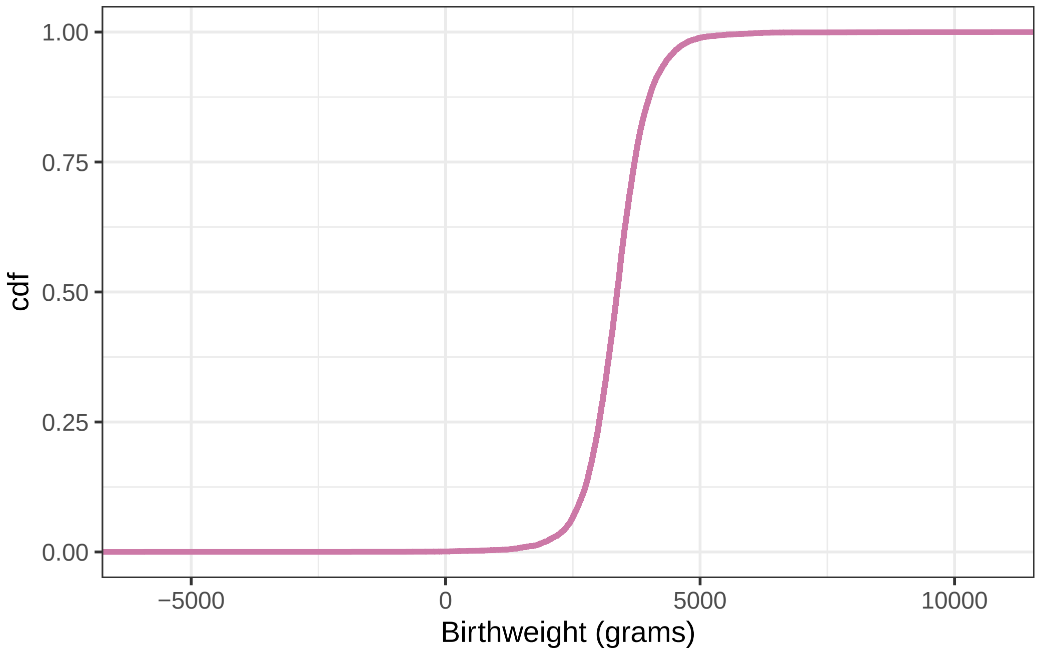
```
n_rep = nrow(posterior)

posterior <- posterior |>
  mutate(y_sim = mu + sigma * rt(n_rep, nu))

posterior |> head(10) |> kbl() |> kable_styling()
```

.chain	.iteration	.draw	mu	sigma	nu	sd	y_sim
1	1	1	3342.971	479.0871	4.288737	655.8148	3183.668
1	2	2	3342.547	455.6609	4.325902	621.4188	3713.447
1	3	3	3379.871	486.4239	4.604530	646.7597	4045.618
1	4	4	3358.972	457.4084	4.224186	630.3621	3199.712
1	5	5	3364.930	471.2112	3.824577	682.2228	3110.418
1	6	6	3365.897	454.1934	3.628184	678.0064	4229.241
1	7	7	3359.967	490.6955	4.884161	638.5536	3816.066
1	8	8	3375.590	462.9417	4.380166	628.0120	3316.929
1	9	9	3359.703	447.5173	3.919852	639.4562	3296.887
1	10	10	3351.785	473.9757	4.294552	648.4344	2803.939

```r
ggplot(posterior, aes(x = y_sim)) +
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["posterior_predict"],
    fill = "white",
    bins = 100
  ) +
  geom_density(linewidth = 1, col = bayes_col["posterior_predict"]) +
  labs(x = "Predicted birthweight (grams)") +
  theme_bw()
```



```r
# plot the cdf
ggplot(posterior, aes(x = y_sim)) +
  stat_ecdf(col = bayes_col["prior_predict"], linewidth = 1) +
  labs(x = "Birthweight (grams)", y = "cdf") +
  theme_bw()
```




```r
quantile(posterior$y_sim, c(0.005, 0.995))
```

```
    0.5%    99.5%
1250.653 5523.626
```

```r
sum(data$birthweight < quantile(posterior$y_sim, 0.005)) /
  length(data$birthweight)
```

```
[1] 0.015
```

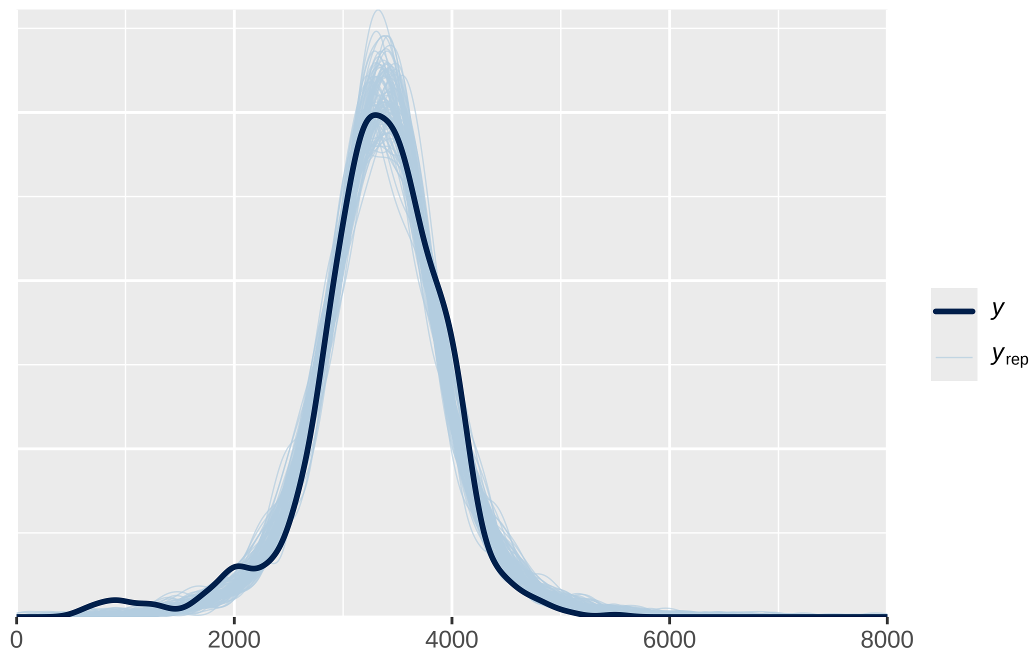
```r
sum(data$birthweight > quantile(posterior$y_sim, 0.995)) /
  length(data$birthweight)
```

```
[1] 0
```

18.1.8 Posterior predictive checking (t likelihood)

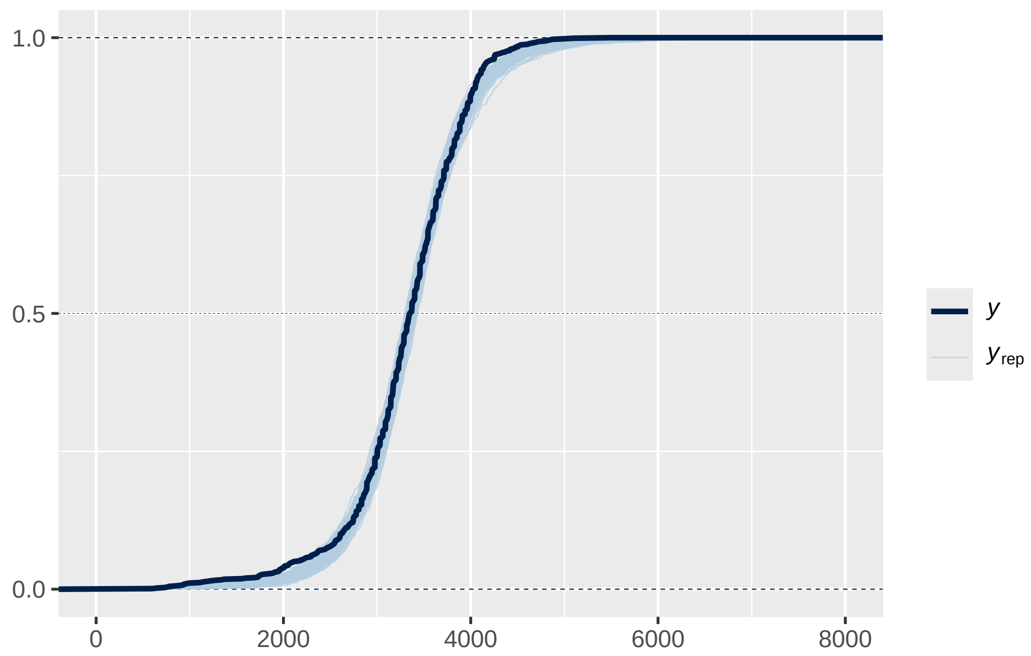
```
pp_check(fit, ndraw = 100) +
  scale_x_continuous(limits = c(0, 8000))
```

```
Warning: Removed 108 rows containing non-finite outside the scale range
(`stat_density()`).
```




```
pp_check(fit, ndraw = 100, type = "ecdf_overlay") +
  scale_x_continuous(limits = c(0, 8000))
```

```
Warning: Removed 94 rows containing non-finite outside the scale range
(`stat_ecdf()`).
```



18.1.9 Default brms priors (t likelihood)

```
fit <- brm(
  data = data,
  family = student,
  birthweight ~ 1,
  iter = 3500,
  warmup = 1000,
  chains = 4,
  refresh = 0
)
```

```
Compiling Stan program...

Start sampling
```

```
prior_summary(fit)
```

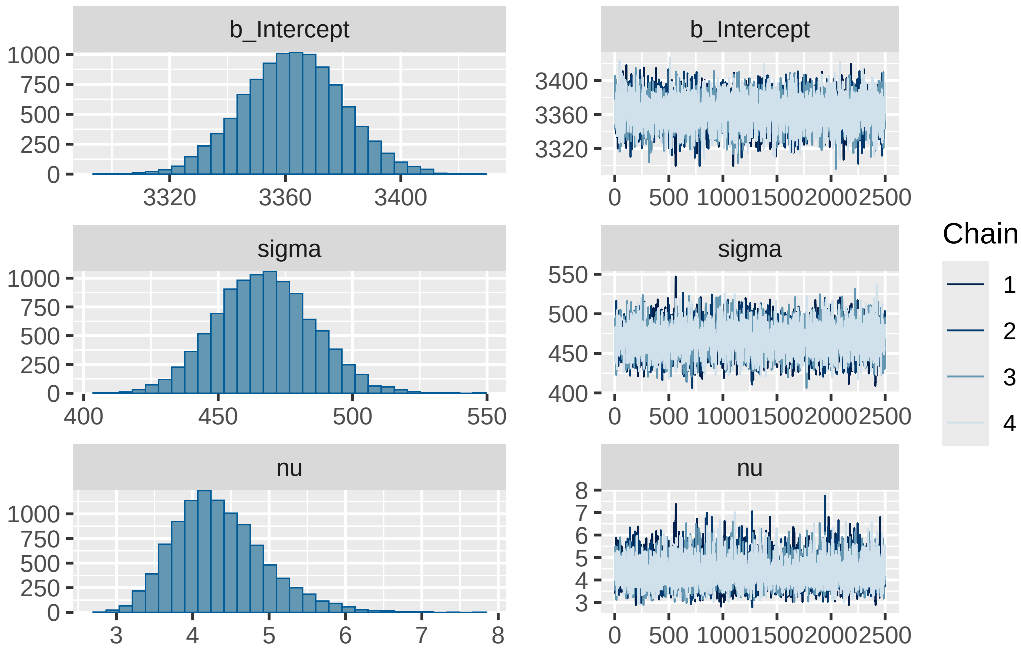
```
                        prior     class coef group resp dpar nlpar lb ub tag
```

```
 student_t(3, 3350.5, 518.9) Intercept
              gamma(2, 0.1)        nu                             1
      student_t(3, 0, 518.9)     sigma                            0
  source
 default
 default
 default
```

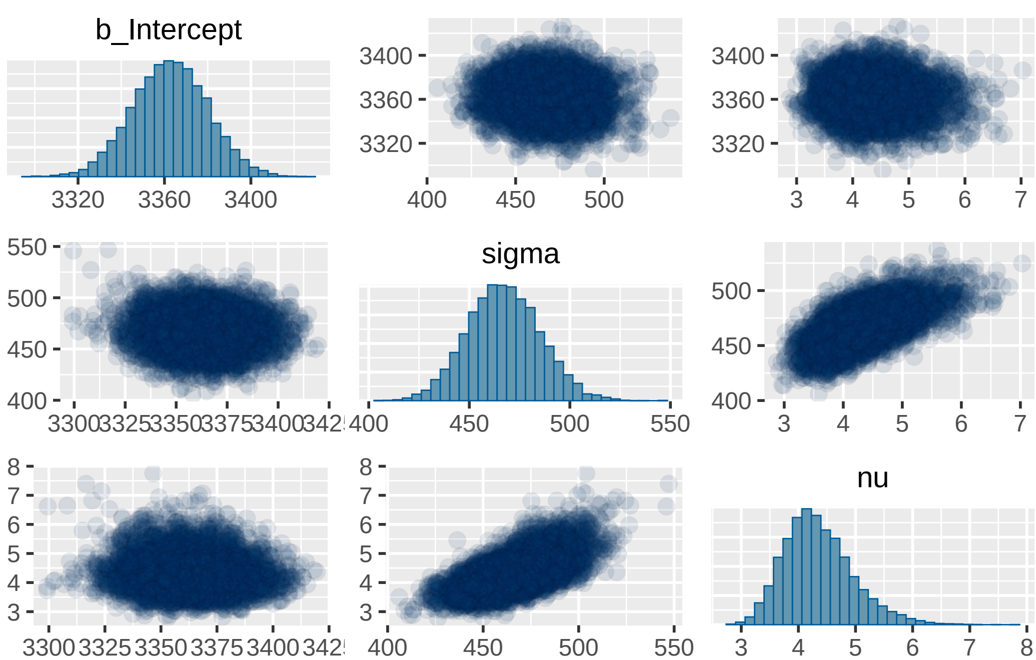
```r
summary(fit)
```

```
 Family: student
  Links: mu = identity
Formula: birthweight ~ 1
   Data: data (Number of observations: 1000)
  Draws: 4 chains, each with iter = 3500; warmup = 1000; thin = 1;
         total post-warmup draws = 10000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept  3362.63     17.35  3328.77  3396.58 1.00     6751     6212

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma   467.30     18.06   433.01   502.86 1.00     5785     6472
nu        4.35      0.60     3.35     5.74 1.00     5531     5559

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit)
```




```r
pairs(fit, off_diag_args = list(alpha = 0.1))
```



```r
pp_check(fit, ndraw = 100) +
  scale_x_continuous(limits = c(0, 8000))
```

```
Warning: Removed 107 rows containing non-finite outside the scale range
(`stat_density()`).
```